

The second-order term of the one-dimensional Taylor tree, with explicit remainder

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Abstract

For $\xi(u) = \xi_0 + cu^m$ we isolate the first two terms of the one-dimensional Taylor tree with an explicit remainder:

$$\begin{aligned} & \left| Z_{\text{chart}} - \frac{1}{2k} n^{-\mu_1} S_{\mu_1}(\xi_0) - \beta c \frac{1}{2k} n^{-\mu_2} S_{\mu_2+1/2}(\xi_0) \right| \\ & \leq \frac{(\beta c)^2}{2} \frac{1}{2k} n^{-(h+2m+1)/(2k)} S_{(h+2m+2k+1)/(2k)}(\xi_0 + |c|b^m) + \text{tail}_0 + |\beta c| \text{tail}_1, \end{aligned}$$

$\mu_1 = (h+1)/(2k)$, $\mu_2 = (h+m+1)/(2k)$, the tails exponentially small. This exhibits the *second* exponent of $\Lambda(h, k)$ with its fluctuation-function coefficient. It rests on a gap-fill $|e^x - 1 - x| \leq \frac{x^2}{2} e^{|x|}$ valid for all real x . Zero `sorry/axiom`.

1 Setting

Unit 26 gives the chart integral as an exact series in p ; unit 27 its leading asymptotic. The next question a reader of `thm:TaylorTree` asks is what the *subleading* term is and how large the error after two terms is. Writing $e^x = 1 + x + R(x)$ with $x = \beta\sqrt{n}u^k cu^m$, the chart integral splits into three pieces: the zeroth- and first-order insertions (known exactly up to tails from units 22/24/25) and a remainder controlled pointwise by $\frac{x^2}{2}e^{|x|}$. On the chart $|x| \leq \beta\sqrt{n}u^k |c|b^m$, so $e^{\text{pop}}e^{|x|}$ is again a population kernel, at the shifted argument $\xi_0 + |c|b^m$, and x^2 contributes $n u^{2k+2m}$: the remainder integral is a fluctuation function at order $(h+2m+2k+1)/(2k)$ times $n \cdot n^{-(h+2m+2k+1)/(2k)} = n^{-(h+2m+1)/(2k)}$.

2 The things formalised

```
theorem abs_exp_sub_one_sub_self_le (x : ℝ) :
  |Real.exp x - 1 - x| ≤ x ^ 2 / 2 * Real.exp |x|

theorem standardIntegral1D_second_order ( n c b : ℝ ) (h k m : ℕ)
  (h : 0 < n) (hn : 1 ≤ n) (hb : 0 < b) (hk : 0 < k) :
  |Z_chart - (1/(2k)) n^(-) S_{ } ( ) - c ((1/(2k)) n^(-) S_{ +1/2} ( ))|
  ( c )^2/2 * ((1/(2k)) n^(-(h+2m+1)/(2k)) S_{(h+2m+2k+1)/(2k)} ( + |c| b^m ))
  + tail + | c | * tail
```

(The Lean statement carries the natural-number casts $\uparrow(h+m)$, $\uparrow(h+2m)$ from the insertion lemma; the tails are the explicit (b, ∞) integrals.)

3 Proof strategy

The exponential bound is the series argument: $e^x - 1 - x = \sum_{p \geq 0} x^{p+2}/(p+2)!$ (`hasSum_nat_add_iff'` at 2), $|x|^{p+2}/(p+2)! \leq \frac{x^2}{2}|x|^p/p!$ from $2p! \leq (p+2)!$, and `hasSum_le` in both directions. For the main theorem, the integrand is decomposed pointwise as $g_0 + g_1 + g_2$ (`funext`, `Real.exp_add`); all three are continuous, hence integrable on the chart (`Continuous.integrableOn_Ioc`), and `integral_add` twice separates them. $\int g_0$ and $\int g_1$ are rewritten as half-line minus tail (`setIntegral_union` on $(0, b] \cup (b, \infty)$) using the unit-22 identity and the $p = 1$ insertion identity. $|\int g_2| \leq \int |g_2|$ (`abs_integral_le_integral_abs`), then `setIntegral_mono_on` against the majorant, enlargement to $(0, \infty)$, and evaluation by the unit-22 identity at $h + 2m + 2k$; the $n \cdot n^{-\alpha}$ merge is one `rpow_add`.

4 Roadblocks and resolutions

- `set` for the local pieces g_0, g_1, g_2 *folded the statement's own tail integral* into $\int g_0$, which would have broken every later syntactic rewrite. Introduce local definitions opaquely — `obtain g, hg : g, g = ... := _, rfl` — so the goal is untouched and the definition is unfolded only where wanted (`simp only [hg]`).
- The idiom `(by rw [h]; exact <multi-line term> : Continuous g)` does not parse when the term wraps lines; give continuity its own `have`.
- `rw [pow_one]` at `hins` fails because the occurrence sits under the integral's binder — use `simp only [pow_one]` at `hins`.
- With an opaque `pop`, `positivity` cannot see $0 < \text{pop } u$; supply `Real.exp_pos` once and thread it.
- The final `gcongr` left the remainder bound stated with the opaque a' against a goal with $\xi_0 + |c|b^m$ unfolded; `rw [← ha']` first.
- Over $\mathbb{N}$, `nlinarith` did not find $2p! \leq (p+2)(p+1)p!$; `Nat.mul_le_mul` with `omega` side goals does.

5 What was Mathlib, what was new

Mathlib: `hasSum_nat_add_iff'`, `hasSum_le`, `abs_integral_le_integral_abs`, `setIntegral_mono_on/mono_set`, `setIntegral_union`, `Continuous.integrableOn_Ioc`. New: the global exponential remainder bound (a lean-common candidate) and the two-term expansion with explicit constant, numerically confirmed (the error tracks $n^{-(h+2m+1)/(2k)}$ with ratio ≈ 0.4 to the bound).

6 Lessons

Use `obtain g, hg : g, g = ...` rather than `set` whenever the statement contains a subterm that the abbreviation would capture: `set` rewrites the goal globally, and a folded statement no longer matches the lemmas built to rewrite it.

7 Follow-ups

An `IsBigO` packaging ($Z - A_0 n^{-\mu_1} - A_1 n^{-\mu_2} = O(n^{-(h+2m+1)/(2k)})$, absorbing the $\sqrt{n}e^{-\varepsilon n}$ tails) and the all-orders version via `standardIntegral1D_monomial_expansion` remain; the latter needs a summability estimate for Γ -weighted exponential coefficients.